Assignment 4 - Software Measurement III - COCOMO II

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Introduction:

In this assignment, we estimated software development cost using the COCOMO II model. This is a parametric model that predicts effort and schedule based on software size and several cost drivers.

And we will be using two sizing methods:

1- Function Points (FP)

2- Source Lines of Code (SLOC)

The estimates are based on the Post-Architecture COCOMO II model, which uses:

- Size (in KSLOC – thousands of SLOC)
- 5 scale factors (SF)**:** PREC, FLEX, RESL, TEAM, PMAT
- 17 cost drivers (effort multipliers, EM)

The main effort equation (Post-Architecture / Early Design form) is: $E = A \times \text{Size}^{B} \times \prod EM_i$

Where:

- E or PM = effort in person-months
- A= constant (2.45 in the manual)
- Size = KSLOC
- B= exponent from scale factors
- $\prod EM_i$= product of the 17 effort multipliers (EAF)

Sizing Method 1: Function Points (FP):

For the first estimation, we used the **Function Points** from the COCOMO II website. Function Points measure the functionality delivered to the user, independent of programming language.

<u>Inputs Used</u>:

- Unadjusted Function Points (UFP): 310
- Language: Basic
- Maintenance: Off
- All Scale Drivers: Nominal
- All Cost Drivers: Nominal
- Cost per Person-Month: $10,000

When all drivers are Nominal, the **Effort Adjustment Factor (EAF)** becomes: $EAF = 1.00$

COCOMO II - Constructive Cost Model

Software Size Sizing Method: Function Points

Unadjusted Function Points: 310 Language: Basic

Software Scale Drivers

Precedentedness	Nominal	Architecture / Risk Resolution	Nominal	Process Maturity	Nominal
Development Flexibility	Nominal	Team Cohesion	Nominal		

Software Cost Drivers

Product		Personnel		Platform	
Required Software Reliability	Nominal	Analyst Capability	Nominal	Time Constraint	Nominal
Data Base Size	Nominal	Programmer Capability	Nominal	Storage Constraint	Nominal
Product Complexity	Nominal	Personnel Continuity	Nominal	Platform Volatility	Nominal
Developed for Reusability	Nominal	Application Experience	Nominal	**Project**	
Documentation Match to Lifecycle Needs	Nominal	Platform Experience	Nominal	Use of Software Tools	Nominal
		Language and Toolset Experience	Nominal	Multisite Development	Nominal
				Required Development Schedule	Nominal

Maintenance Off

Software Labor Rates

Cost per Person-Month (Dollars): 10000

Calculate

Results

Software Development (Elaboration and Construction) Staffing Profile

Effort = 78.6 Person-months
Schedule = 14.7 Months
Cost = $785694

Total Equivalent Size = 19840 SLOC
Effort Adjustment Factor (EAF) = 1.00

Acquisition Phase Distribution

Phase	Effort (Person-months)	Schedule (Months)	Average Staff	Cost (Dollars)
Inception	4.7	1.8	2.6	$47142
Elaboration	18.9	5.5	3.4	$188567
Construction	59.7	9.2	6.5	$597128
Transition	9.4	1.8	5.1	$94283

Software Effort Distribution for RUP/MBASE (Person-Months)

Phase/Activity	Inception	Elaboration	Construction	Transition
Management	0.7	2.3	6.0	1.3
Environment/CM	0.5	1.5	3.0	0.5
Requirements	1.8	3.4	4.8	0.4
Design	0.9	6.8	9.6	0.4
Implementation	0.4	2.5	20.3	1.8
Assessment	0.4	1.9	14.3	2.3
Deployment	0.1	0.6	1.8	2.8

The tool outputs:

- Total Equivalent Size = 19,840 SLOC
- Effort E = 78.6 person-months
- Schedule TDEV = 14.7 months
- Cost = $785,694.00
- EAF = 1.00

The FP method showing that based on 310 function points, the project is almost **20,000 SLOC** and will cost about **$785k** and take about **15 months**.

The results also show that the Construction phase has the highest effort (59.7PM), which is typical since most coding and integration happen there.

Sizing Method 2: Source Lines of Code (SLOC)

SLOC is more technical and precise once the design is detailed because it counts the logical source statements in the system. COCOMO II uses SEI's definition checklist to ensure consistent SLOC counting across languages.

Inputs Used:

- New Sloc: 20000
- Reused SLOC: 1000
- Modified SLOC: 1000
- % code Modified: 5%
- Maintenance: Off
- Cost per Person-Month: $10,000
- All Scale Drivers: Nominal
- All Cost Drivers: Nominal

COCOMO II - Constructive Cost Model

Software Size Sizing Method: Source Lines of Code

	SLOC	% Design Modified	% Code Modified	% Integration Required	Assessment and Assimilation (0% - 8%)	Software Understanding (0% - 50%)	Unfamiliarity (0-1)
New	20000						
Reused	1000	0	0	0	0		
Modified	1000	0	5	0	0	0	0

Software Scale Drivers

Precedentedness	Nominal	Architecture / Risk Resolution	Nominal	Process Maturity	Nominal
Development Flexibility	Nominal	Team Cohesion	Nominal		

Software Cost Drivers

Product		Personnel		Platform	
Required Software Reliability	Nominal	Analyst Capability	Nominal	Time Constraint	Nominal
Data Base Size	Nominal	Programmer Capability	Nominal	Storage Constraint	Nominal
Product Complexity	Nominal	Personnel Continuity	Nominal	Platform Volatility	Nominal
Developed for Reusability	Nominal	Application Experience	Nominal	**Project**	
Documentation Match to Lifecycle Needs	Nominal	Platform Experience	Nominal	Use of Software Tools	Nominal
		Language and Toolset Experience	Nominal	Multisite Development	Nominal
				Required Development Schedule	Nominal

Maintenance Off

Software Labor Rates

Cost per Person-Month (Dollars): 10000

Calculate

Results

Software Development (Elaboration and Construction) Staffing Profile



Effort = 79.3 Person-months
Schedule = 14.7 Months
Cost = $793319

Total Equivalent Size = 20015 SLOC
Effort Adjustment Factor (EAF) = 1.00

Acquisition Phase Distribution

Phase	Effort (Person-months)	Schedule (Months)	Average Staff	Cost (Dollars)
Inception	4.8	1.8	2.6	$47599
Elaboration	19.0	5.5	3.4	$190397
Construction	60.3	9.2	6.5	$602923
Transition	9.5	1.8	5.2	$95198

Software Effort Distribution for RUP/MBASE (Person-Months)

Phase/Activity	Inception	Elaboration	Construction	Transition
Management	0.7	2.3	6.0	1.3
Environment/CM	0.5	1.5	3.0	0.5
Requirements	1.8	3.4	4.8	0.4
Design	0.9	6.9	9.6	0.4
Implementation	0.4	2.5	20.5	1.8
Assessment	0.4	1.9	14.5	2.3
Deployment	0.1	0.6	1.8	2.9

The tool outputs:

- Total Equivalent Size = 20,015 SLOC
- Effort E = 79.3 person-months
- Schedule TDEV = 14.7 months
- Cost = $793,179.00
- EAF = 1.00

The SLOC method shows that based on 20,000 new SLOC, plus 1,000 reused SLOC and 1,000 modified SLOC (with 5% of the modified code actually changed), the project has an effective size of about **20,015** SLOC. With all cost and scale drivers set to Nominal, the estimated effort is about 79.3 person-months, the schedule is about **15 months**, and the total development cost is around **$793k**.

The results also show that the Construction phase again has the highest effort **(**60.3 PM**).**

The impact of different configurable parameters/Factors on the software estimation cost:

The COCOMO II model includes several configurable parameters on the estimation website that significantly influence the final effort, schedule, and total project cost. These parameters determine how work is distributed across the software lifecycle, how many development cycles are needed, and whether additional maintenance work is required.

1. **Phase Distribution:**
   The distribution parameter controls how the tool allocates effort across the four major lifecycle phases: Inception, Elaboration, Construction, and Transition. Since Construction is the most effort-heavy phase (coding, integration, and testing), increasing the percentage of effort allocated to this phase raises the total number of person-months and therefore increases cost. Changing distribution shifts workload between phases, which directly impacts staffing needs and schedule.

2. **Number of Iterations:**
   Iterations represent how many development cycles the team will go through (for example, multiple increments or sprints). More iterations introduce repeated planning, design, coding, and testing activities. As a result, total effort and schedule increase, because each iteration adds overhead and rework. Fewer iterations reduce cost by lowering the repeated effort associated with each cycle.

3. **Maintenance Type:**
   Turning maintenance ON adds additional effort to cover post-deployment updates, bug fixes, enhancements, and long-term support. Maintenance usually adds a large percentage of total effort and increases the schedule beyond initial development. When maintenance is included, cost rises because more person-months and staff allocation are required after release.

4. **Cost Drivers and Scale Factors:**
   cost drivers such as product complexity, required reliability, platform volatility, or personnel capability all modify the Effort Adjustment Factor (EAF). Higher ratings (High or Very High) significantly increase cost because they represent a more difficult or risk-heavy project. Similarly, scale factors influence the exponent on effort; higher scale factors lead to exponential growth in effort.

5. **Labor Rate (Cost per Person-Month):**
   This parameter directly multiplies the effort. A higher labor rate immediately increases the total cost, even if effort stays the same.

Conclusion:

Both the Function Point and SLOC methods provided very similar results in terms of effort, schedule, and cost, which shows consistency in the COCOMO II model. FP estimated 78.6 PM and $785k, while SLOC estimated 79.3 PM and $793k. The analysis of configurable parameters demonstrated that factors such as distribution, iterations, maintenance, cost drivers, and scale factors can significantly change the final cost estimate. Overall, the COCOMO II tool helps illustrate how different project assumptions and lifecycle decisions directly influence the total development effort and cost.